

Toward a Mechanistic DEB Model for *Aporrectodea caliginosa* - Linking Food Availability, Organic Matter Quality, and Energy Assimilation

Supplementary information

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1 Earthworm culture

1.1 Breeding and experiment soil

The soil used in the culture of earthworms is collected from the top 10 cm of a meadow in Versailles called “Les Closeaux”. It is then dried before being milled at < 2 mm.

Table 1.1: Main characteristics of the breeding soil from Versailles. Table from Bart et al. (2017).

Characteristics	Value
Clay (< 2 μm , g/kg)	226.0
Fine silt (2-20 μm , g/kg)	174.1
Coarse silt (20-50 μm , g/kg)	298.9
Fine sand (50-200 μm , g/kg)	239.1
Coarse sand (200-2000 μm , g/kg)	47.9
CaCO_3 total (g/kg)	23.3
Organic matter (g/kg)	32.6
P_2O_5 (g/kg)	0.1
Organic carbon (g/kg)	18.9
Total nitrogen (N) (g/kg)	1.5
C/N	12.5
pH	7.5
Total Cu (mg/kg)	25.2

1.2 DNA barcoding

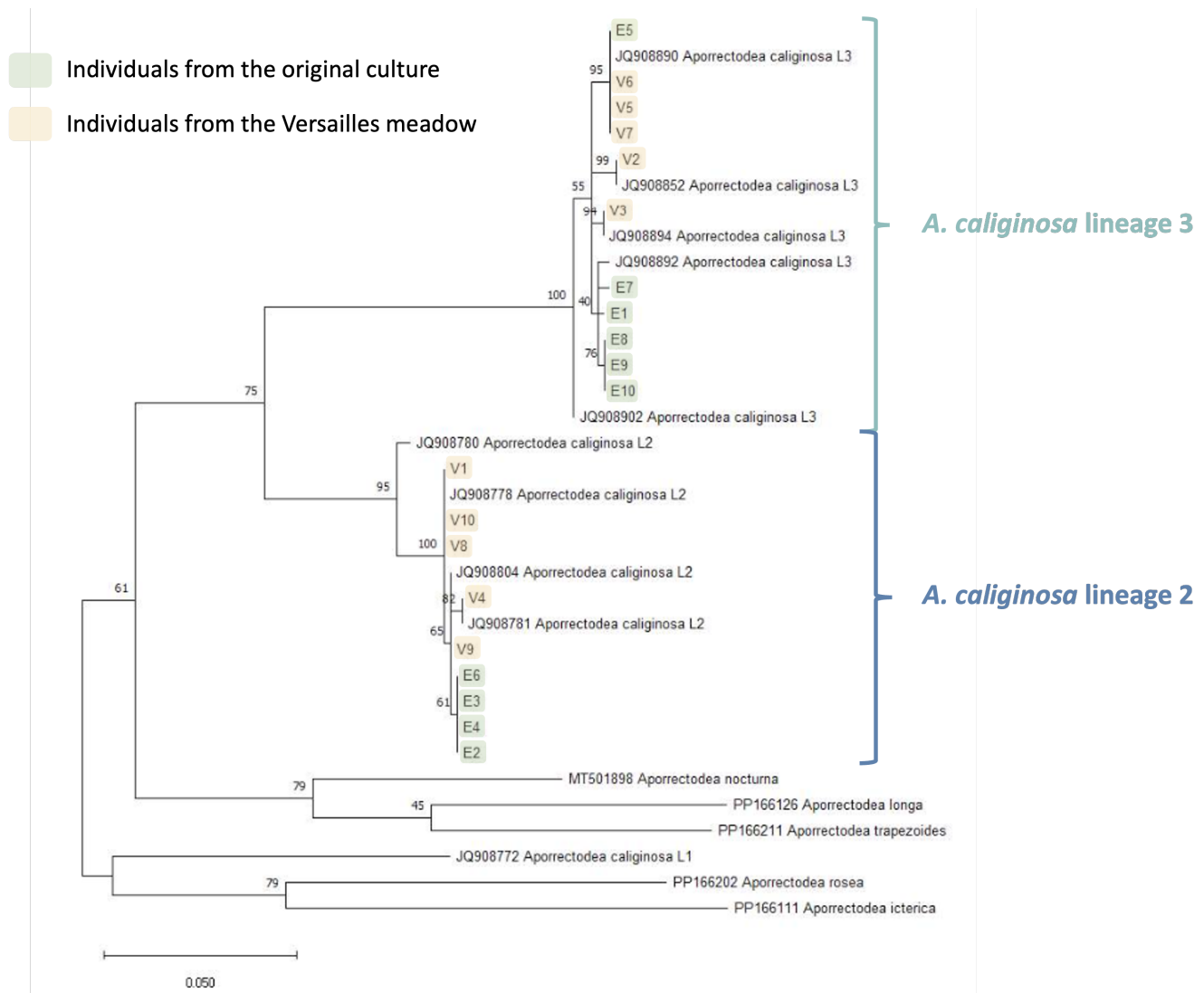


Figure 1.1: Results of the DNA barcoding of 10 individuals from the culture (highlighted in yellow)

The phylogenetic tree was realised thanks to maximum likelihood bootstrap method (1000 simulations) with the Tamura-Nei model (nucleotides). Bootstrap frequency is indicated on each branch.

Aporrectodea caliginosa is a cryptic species complex comprising three lineages. Our earthworm culture is known to include individuals from two closely related lineages (2 and 3). While these lineages could potentially exhibit differences in sensitivity, previous research on *Gammarus roeselii* has shown that tolerance within cryptic species complexes is not always lineage dependent (Kabus et al., 2024).

2 Experiments

The Table 2.1 presents the characteristics of each experiment used to calibrate the DEB model.

Table 2.1: Summary of the experimental conditions and measured endpoints for each experiment used to calibrate the DEB model. The numbering for Bart (2019) follows that of the original publication. Control individuals from Bart et al. (2020) include those from the XP2 experiment reported in Bart (2019).

Reference	Experiment name	Soil in cosm	Food (g/ind/14d)	Density (#/cosm)	Temperature (°C)
Bart (2019)	XP1_1	Closeaux soil (400 g, WHC 70%)	3	1	15
	X1_2	Closeaux soil (400 g, WHC 70%)	3	1 then 2 from day 112, 3 from day 210 and 6 from day 294	15
	XP_3	Closeaux soil (400 g, WHC 70%)	0 but 3 at days 70, 84 and 140	1	15
	XP4_1	Closeaux soil (400 g, WHC 70%)	1	1	15
	XP4_2	Closeaux soil (400 g, WHC 70%)	1	1 then 2 from day 140, 3 from day 224 and 6 from day 322	15
	XP5	Closeaux soil (400 g, WHC 70%)	1.5	1	15
Bart et al. (2020)	Control individuals	Closeaux soil (400 g, WHC 70%)	3	1	15

Reference	Experiment name	Soil in cosm	Food (g/ind/14d)	Density (#/cosm)	Temperature (°C)
Present paper	D1	Closeaux soil (200 g, WHC 70%)	3	1	18
	D2	Closeaux soil (200 g, WHC 70%)	1.5	2	18
	D2b	Closeaux soil (200 g, WHC 70%)	3	2	18
	D5	Closeaux soil (200 g, WHC 70%)	0.6	5	18
	D7	Closeaux soil (200 g, WHC 70%)	0.4	7	18

3 Models

3.1 Priors used

Priors are the same between models :

- `Distrib (Fm, TruncNormal, 0.1, 0.2, 0.0001, 0.8);`
- `Distrib (pAm, TruncNormal_cv, 100, 0.2, 40, 2000);`
- `Distrib (r_pAm_pM, TruncNormal_cv, 0.8, 0.1, 0.1, 2);`
- `Distrib (v, TruncNormal_cv, 0.18505, 0.1, 0.001, 2);`
- `Distrib (kap, Uniform, 0, 1);`
- `Distrib (kapJ, Uniform, 0, 1);`
- `Distrib (kapA, Uniform, 0, 1);`
- `Distrib (Ehp, TruncNormal_cv, 100, 0.2, 20, 300);`
- `Distrib (E_coc, TruncNormal_cv, 30, 0.2, 1, 300);`
- `Distrib (rOM_ClxHorse, LogUniform, 0.00001, 0.3);`
- `Distrib (K, Uniform, 0.0001, 1);`
- `Distrib (dv, TruncNormal_cv, 1, 0.05, 0, 4);`
- `Distrib (wE, LogUniform, 1, 1000);`
- `Distrib (de, LogUniform, 0.000000001, 0.1);`
- `Distrib (tau, TruncNormal_cv, 7, 0.5, 0, 50);`

Log-Likelihood calculations :

- `Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);`
- `Likelihood (Weight, Normal, Prediction(Weight), Sigma_W);`
- `Likelihood (Reproduction, Poisson, Prediction(Reproduction));`

3.2 Model A_FLYw

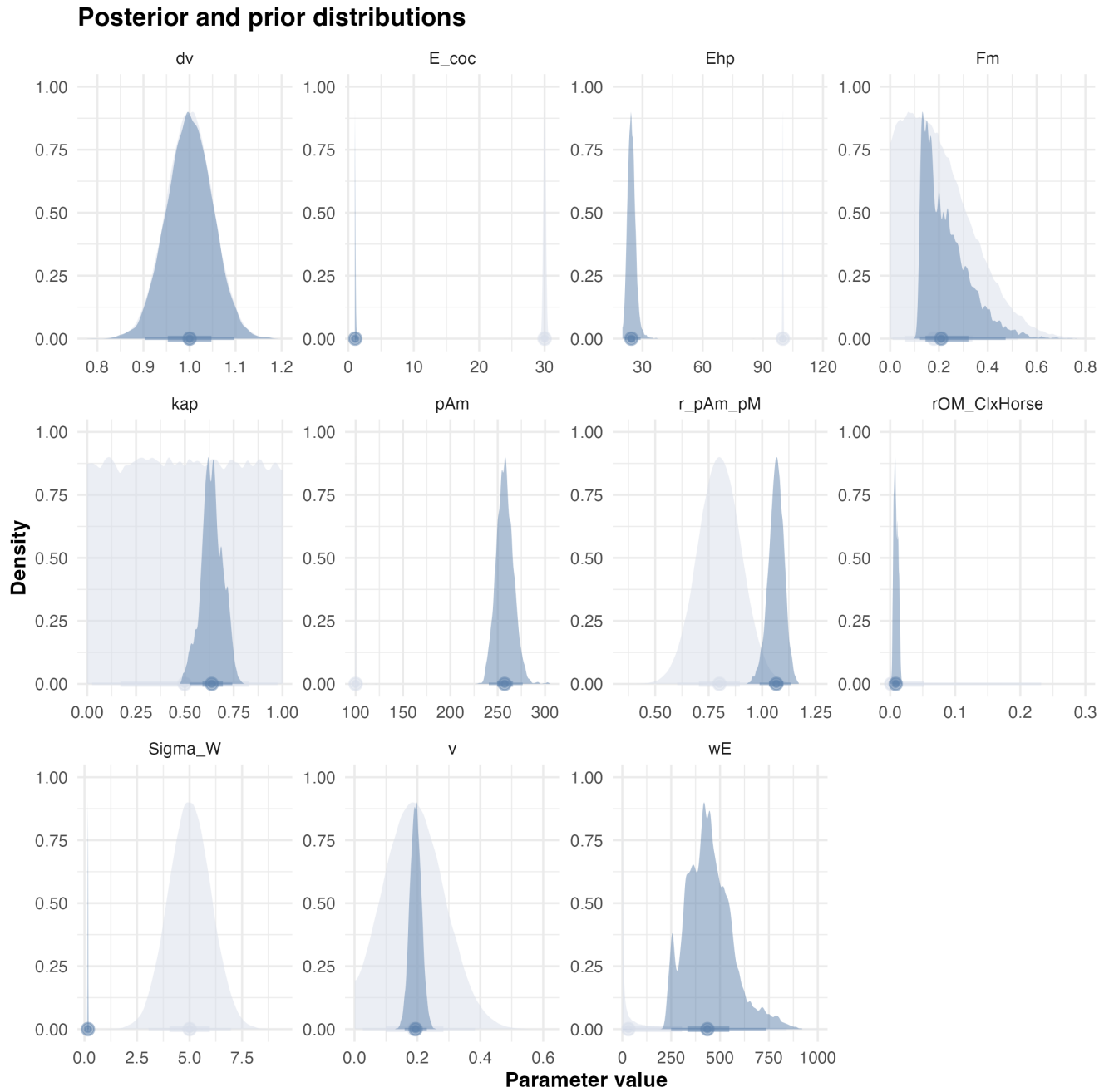


Figure 3.1: Posteriors.

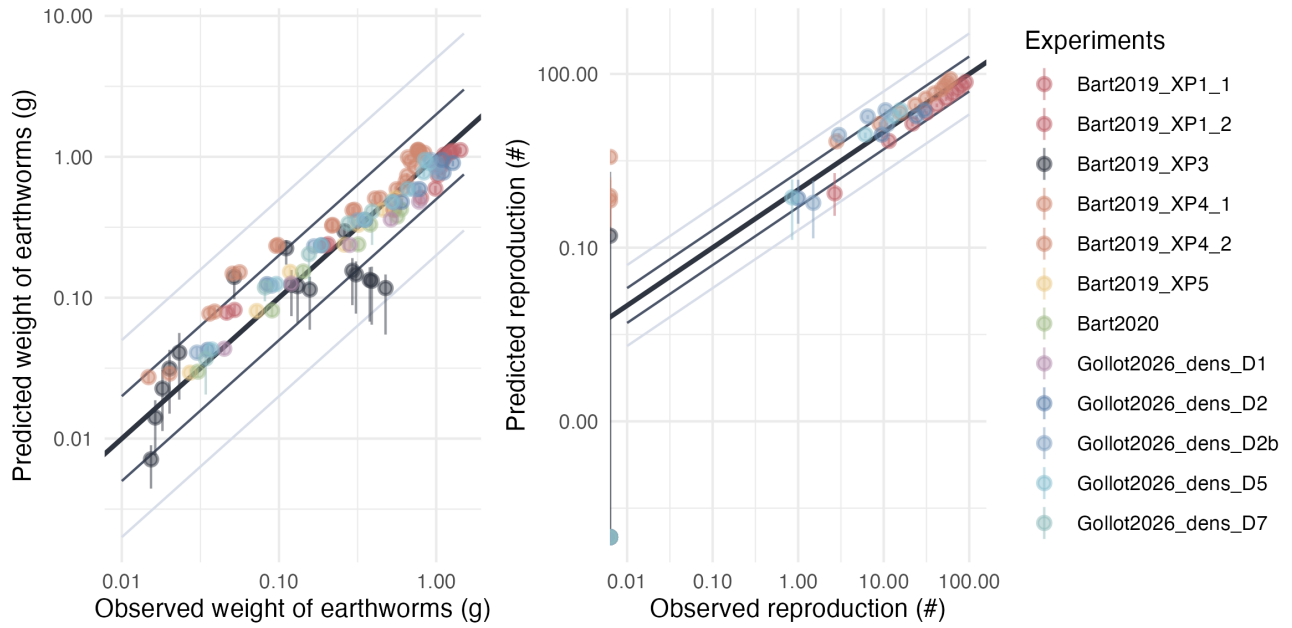


Figure 3.2: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

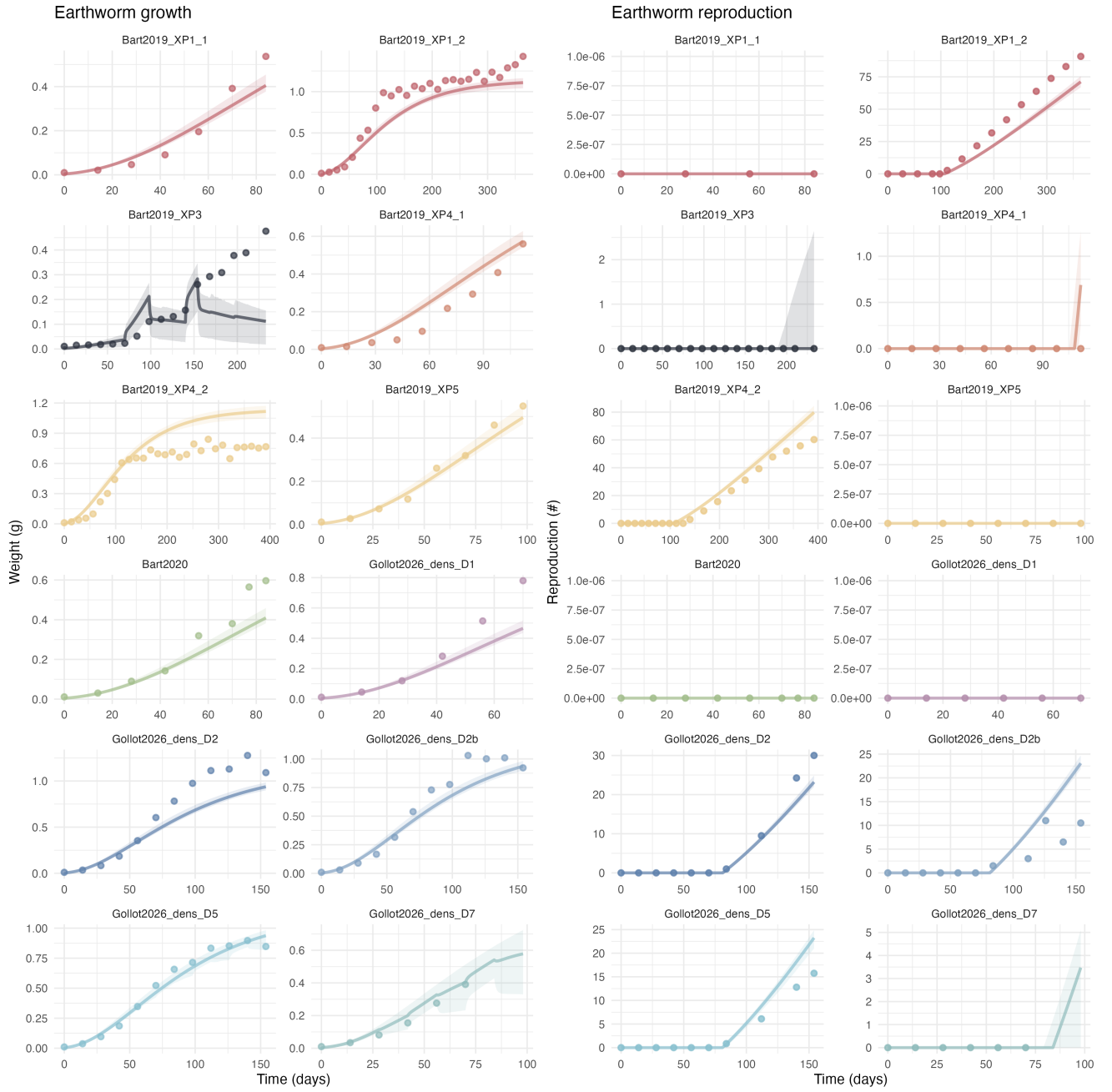


Figure 3.3: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

3.3 Model B_FLYe

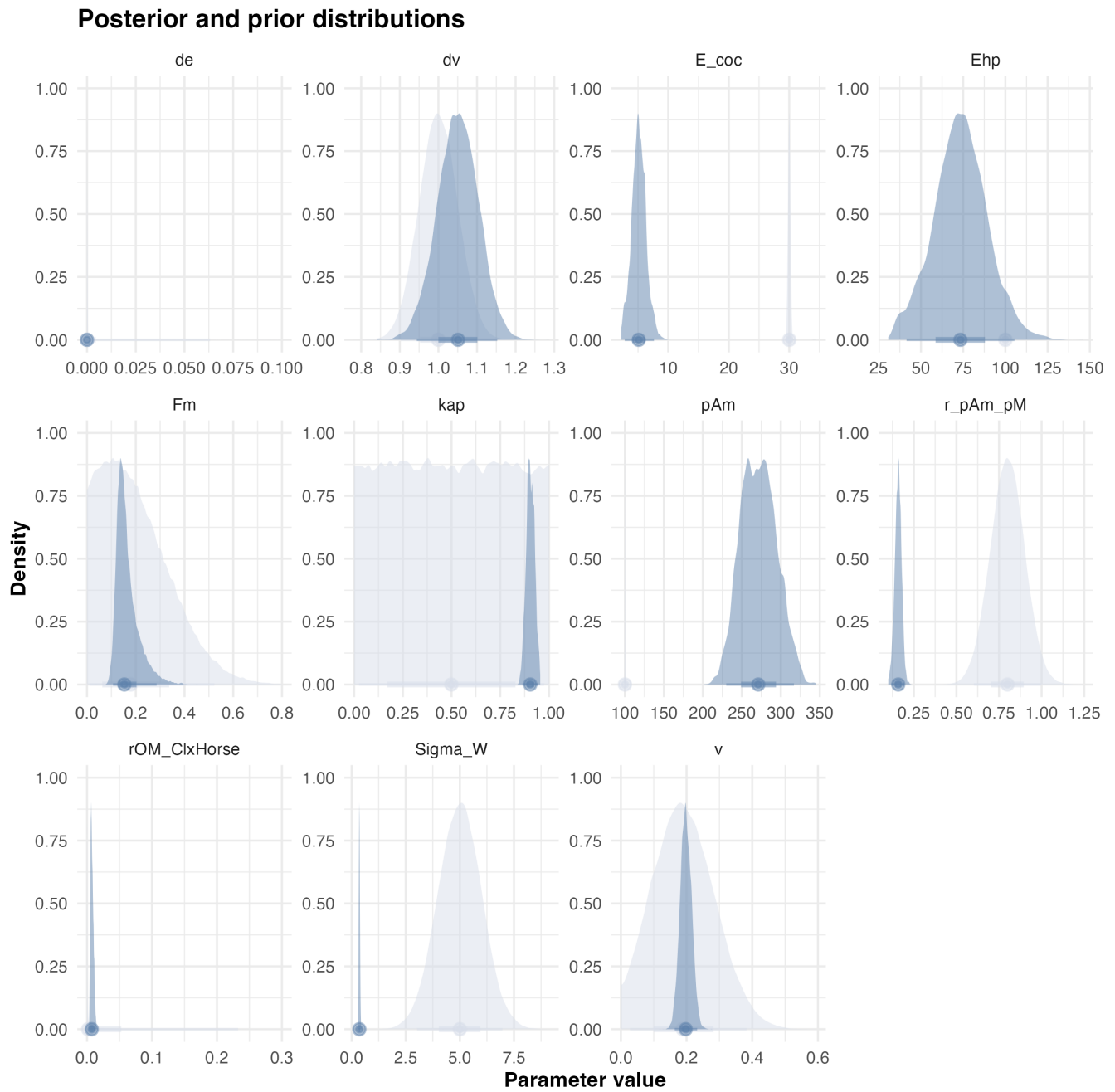


Figure 3.4: Posteriors.

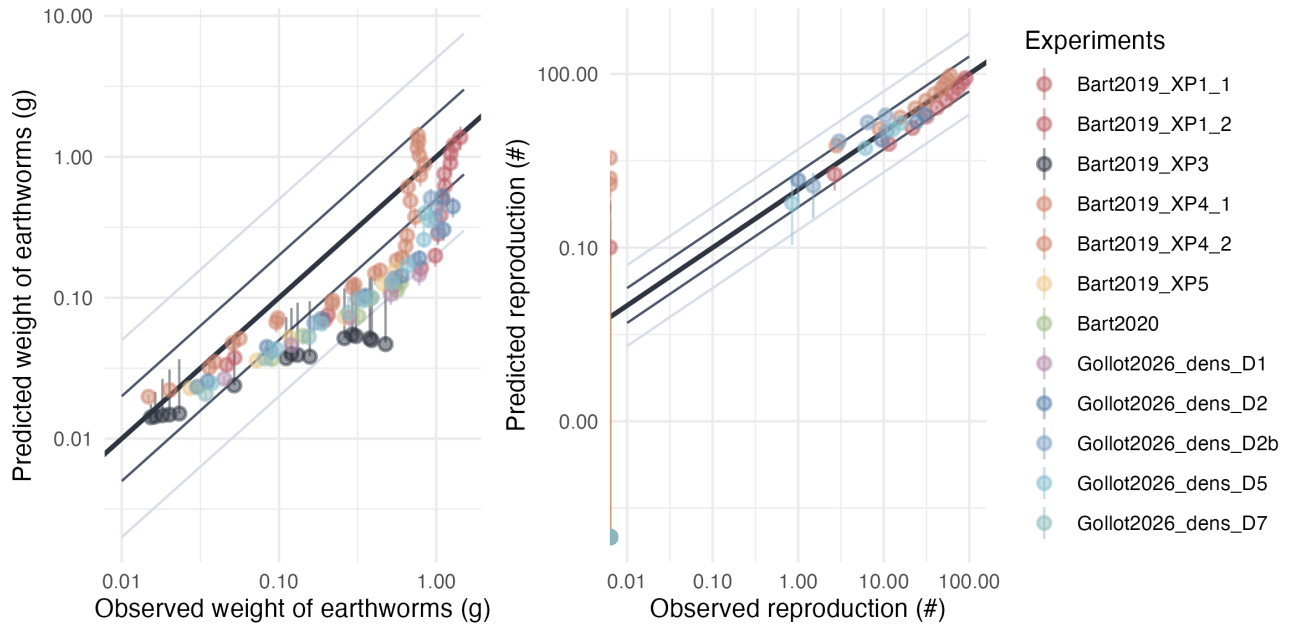


Figure 3.5: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

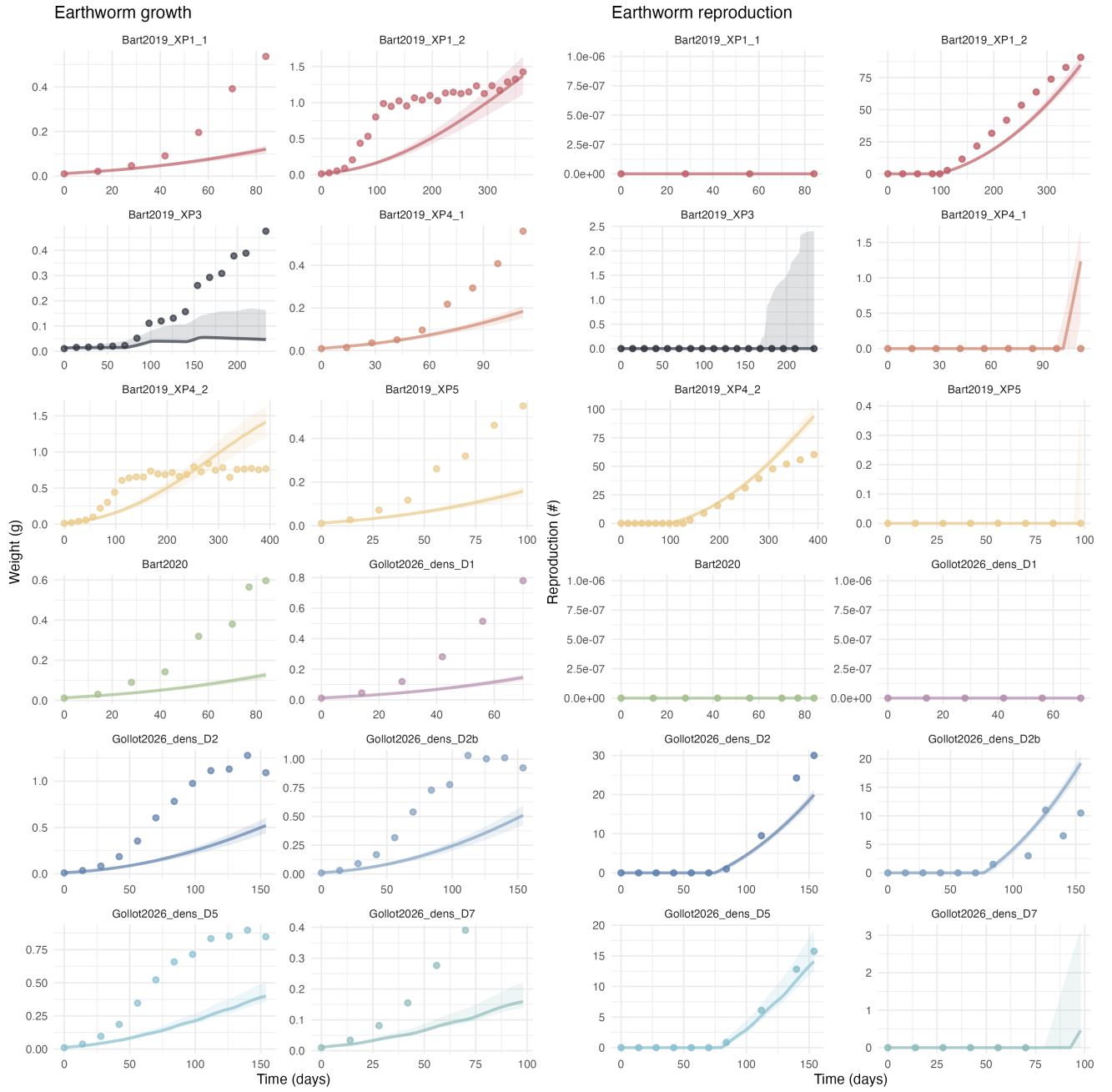


Figure 3.6: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

3.4 Model C_CWYe

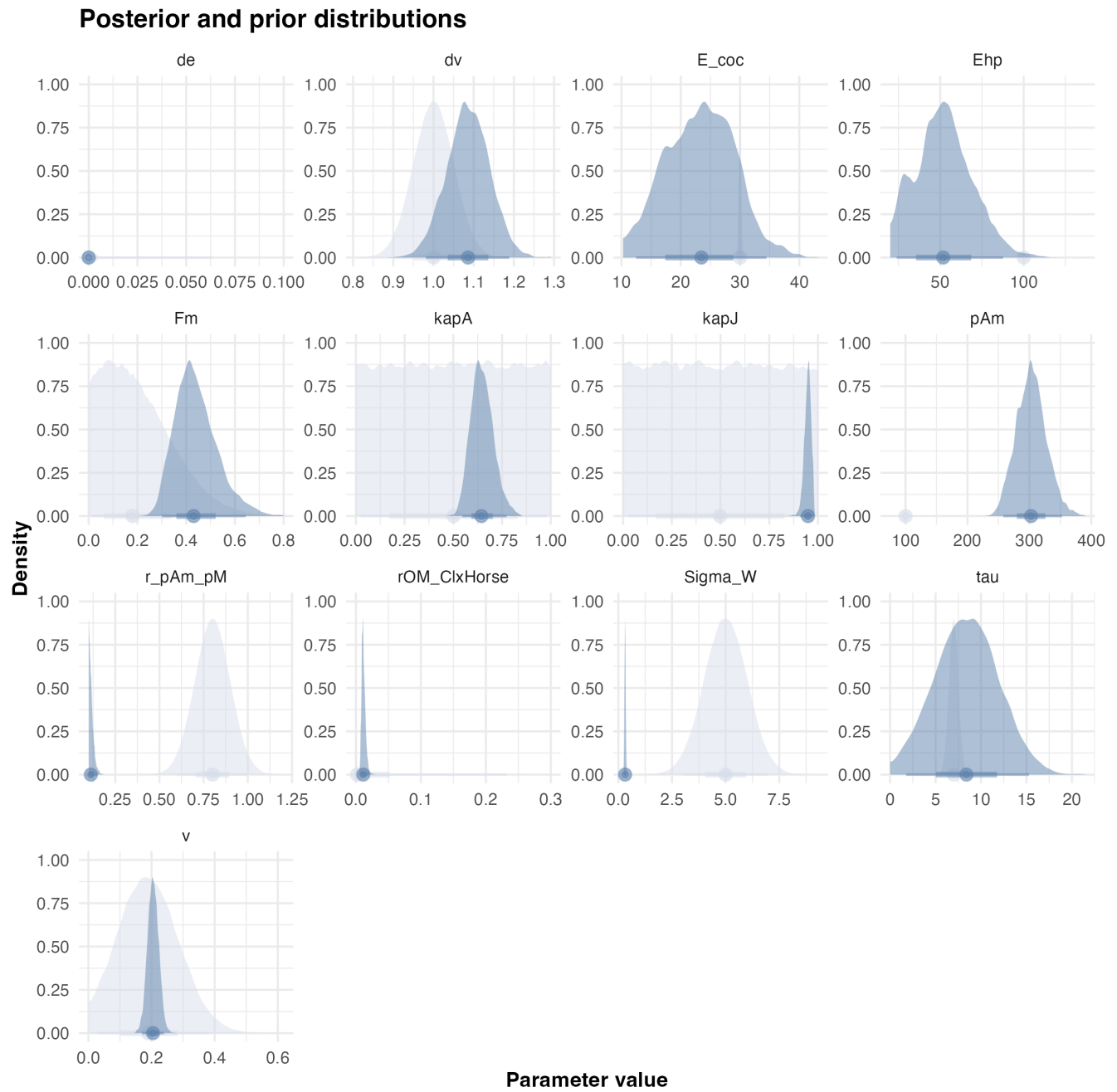


Figure 3.7: Posteriors.

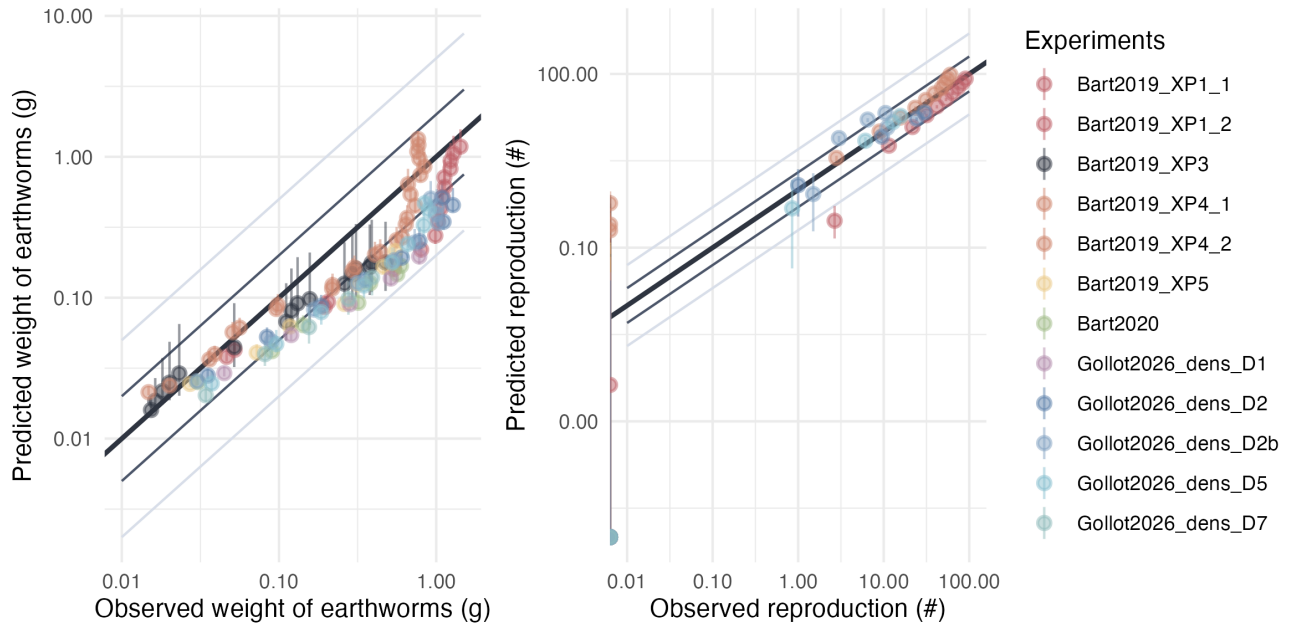


Figure 3.8: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

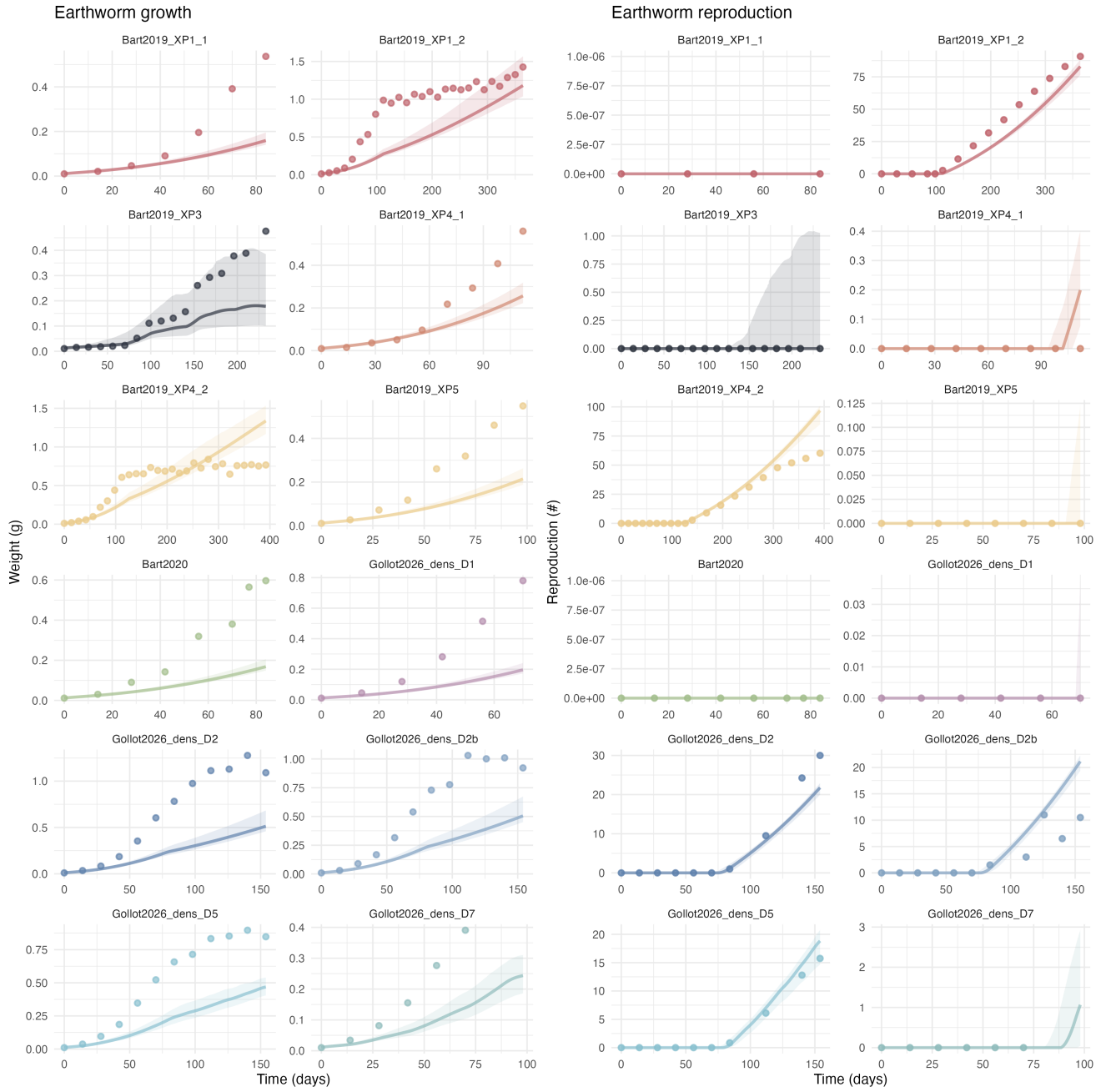


Figure 3.9: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

3.5 Model D_CWKe

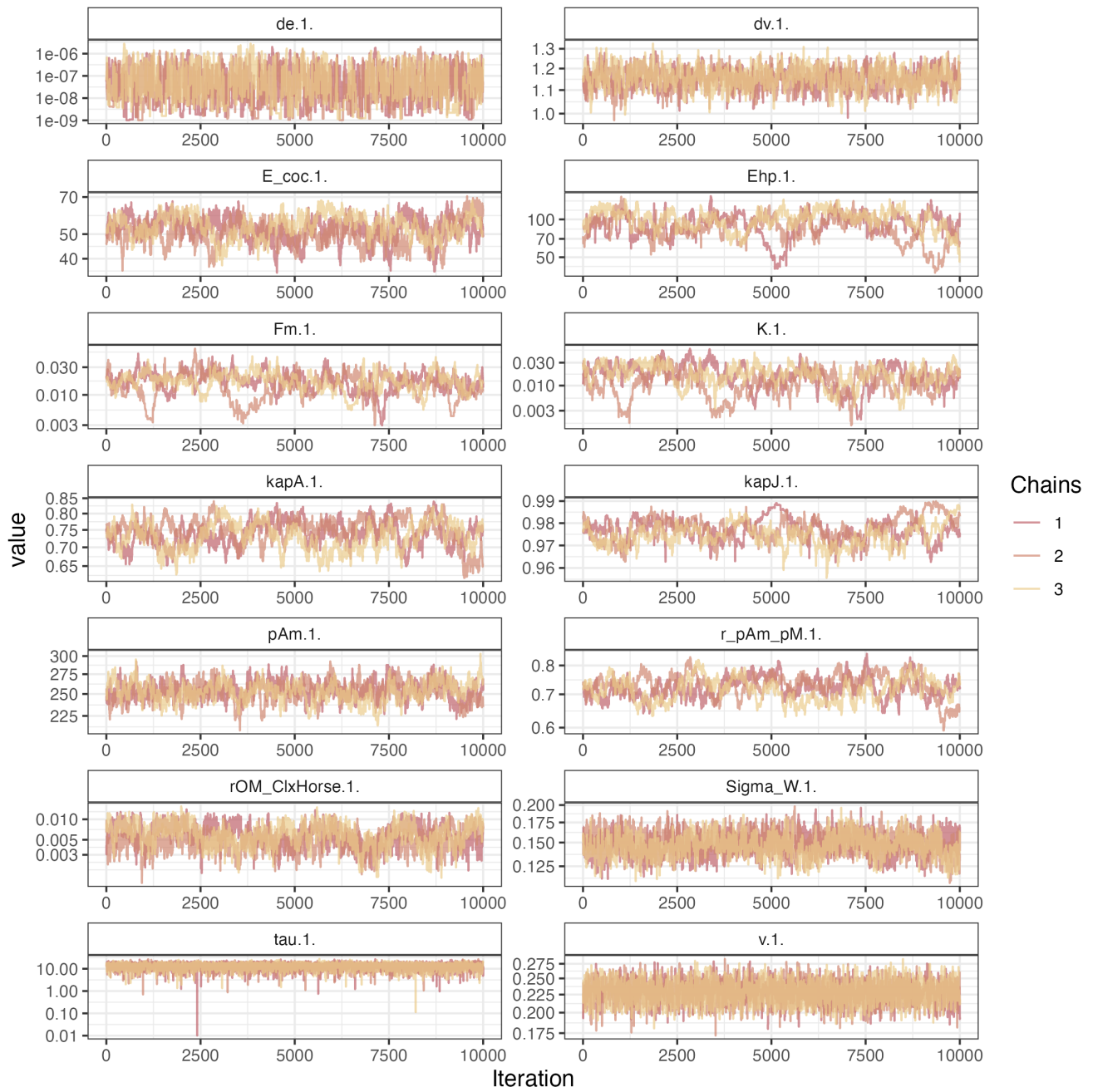


Figure 3.10: Chains.

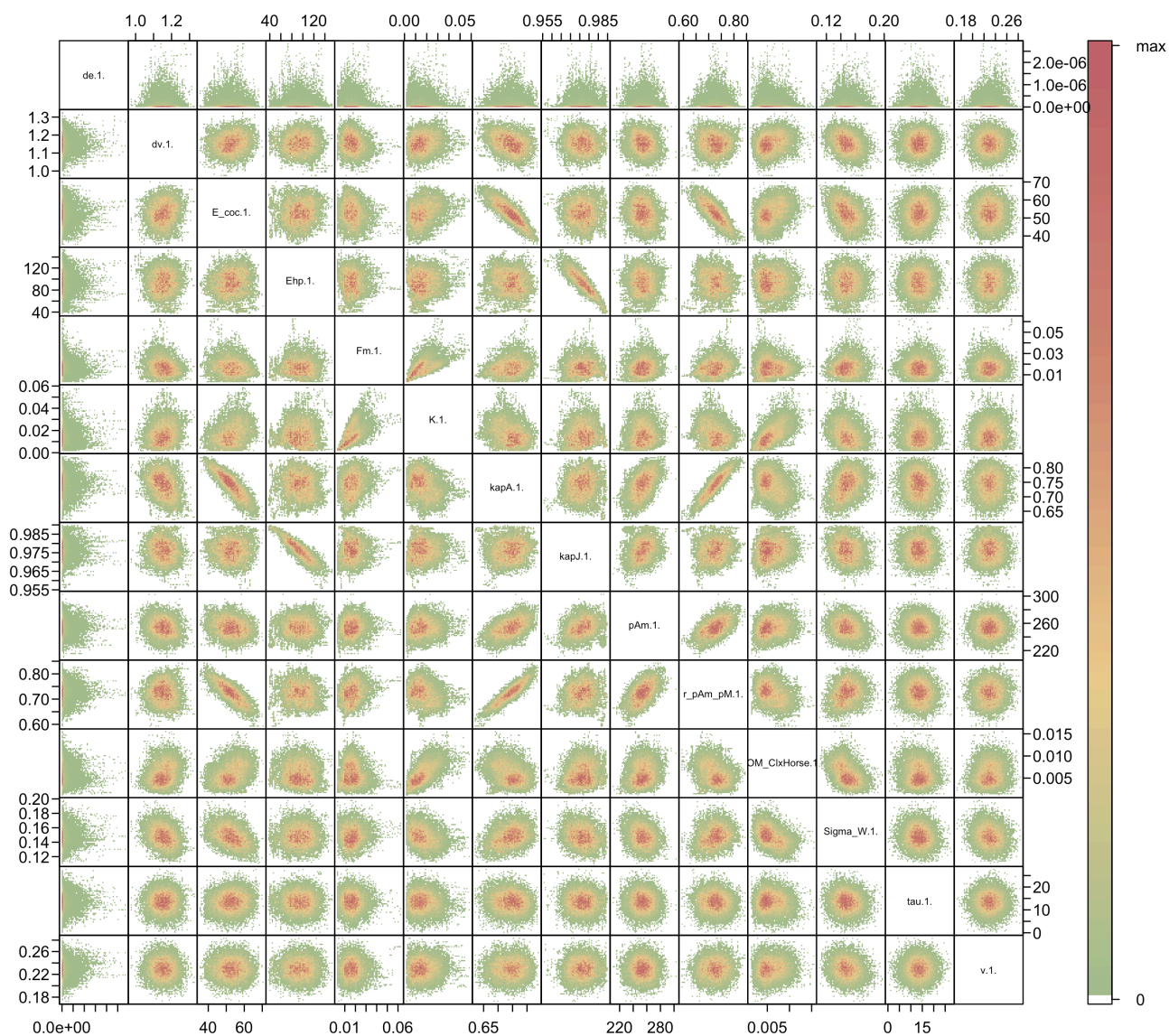


Figure 3.11: Chain correlations.

Posterior and prior distributions

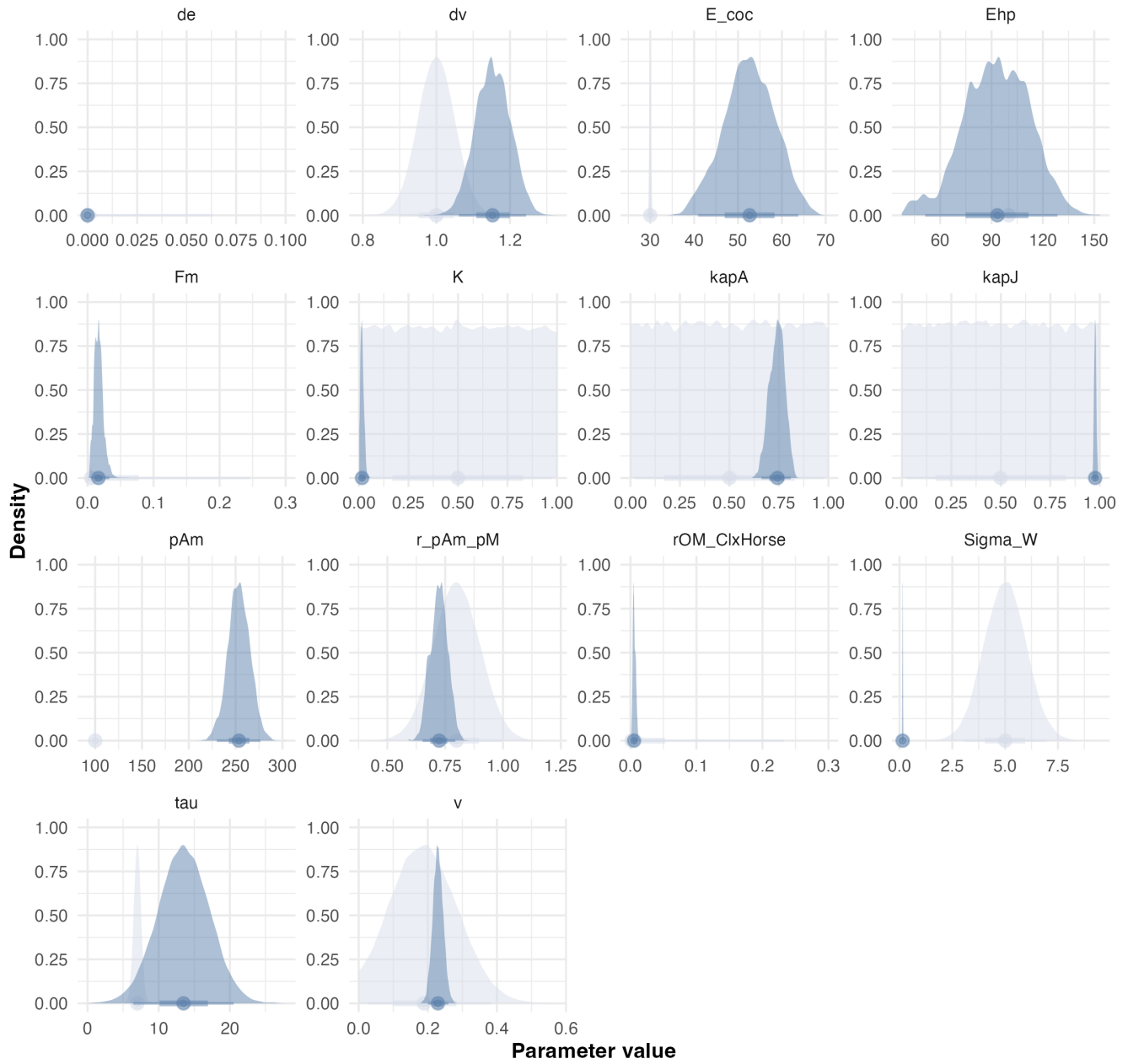


Figure 3.12: Posteriors.

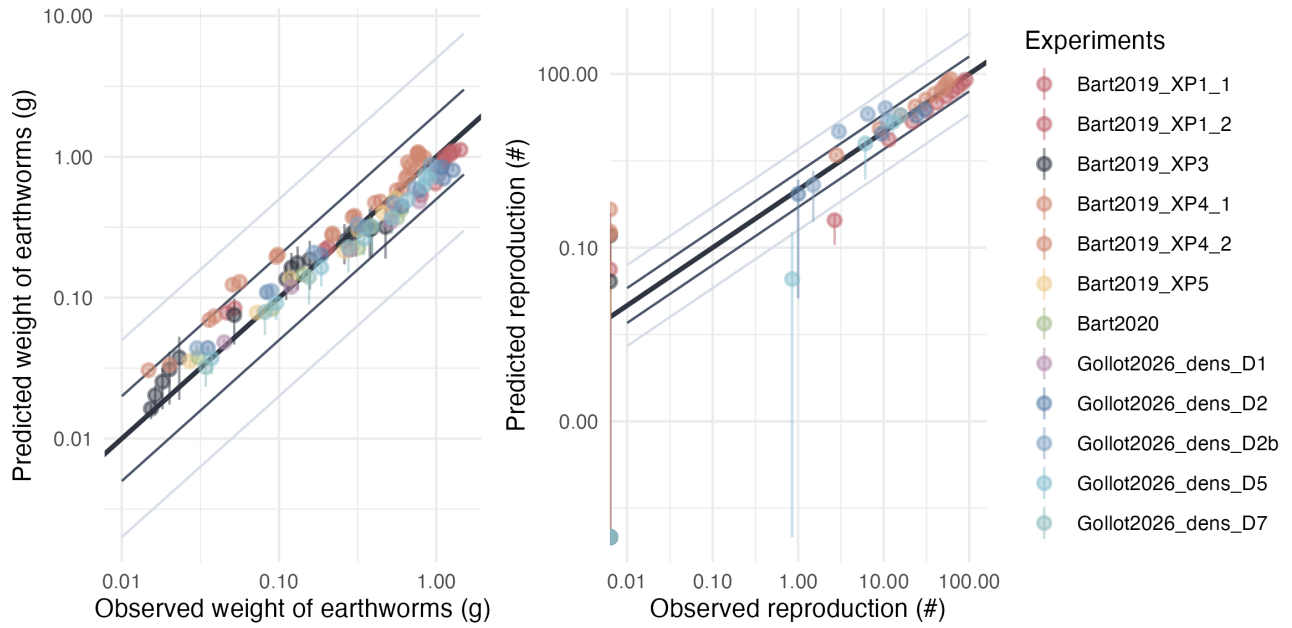


Figure 3.13: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

3.6 Model E_CLYe

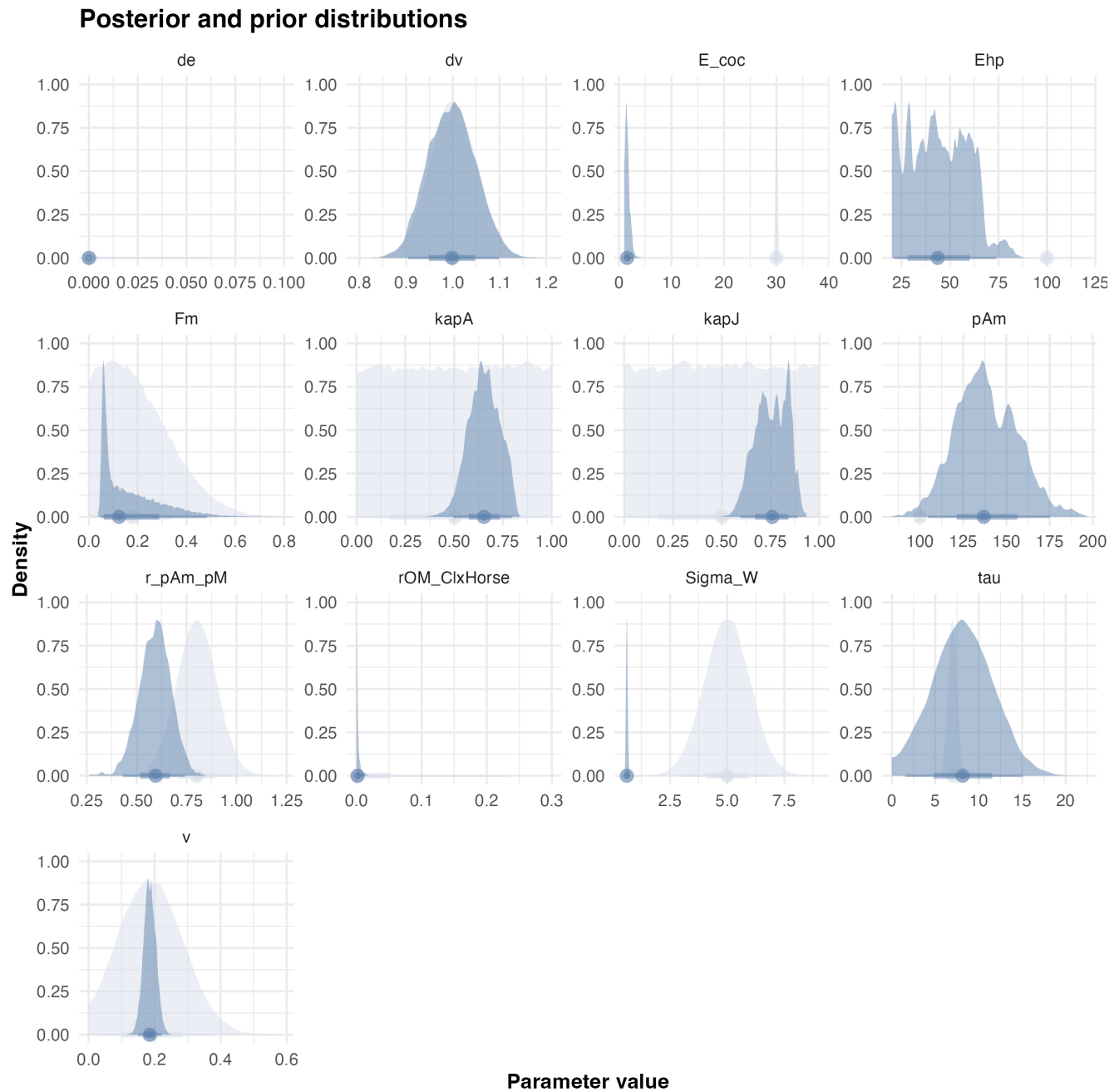


Figure 3.14: Posteriors.

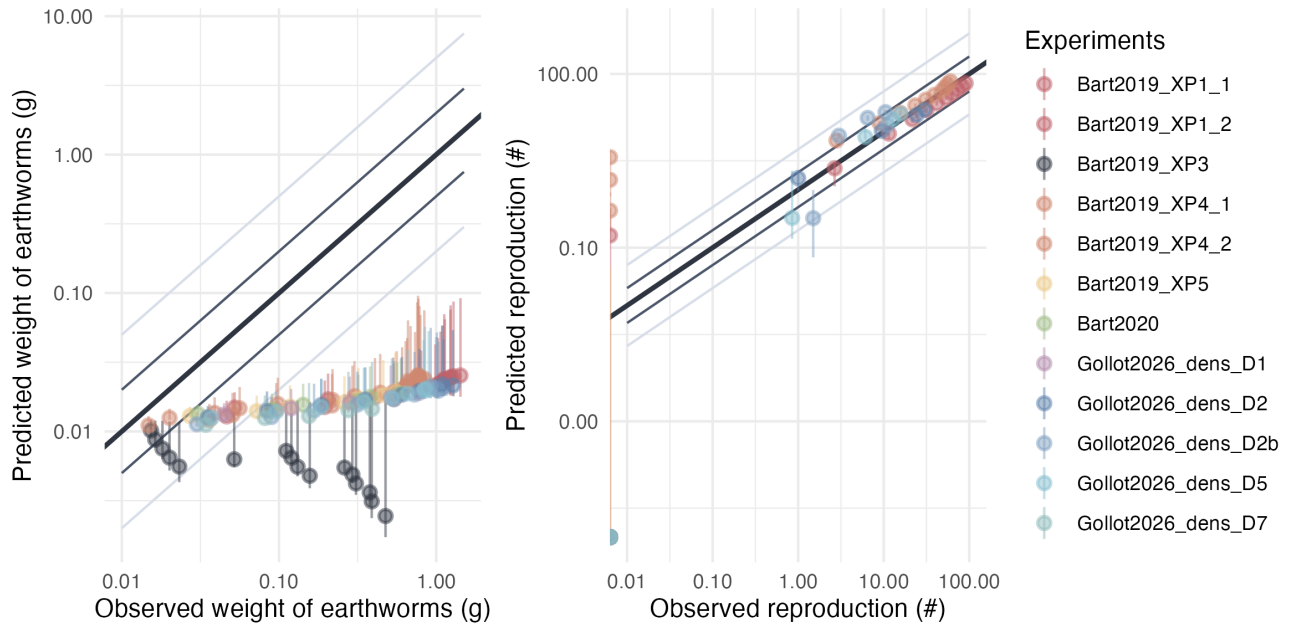


Figure 3.15: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

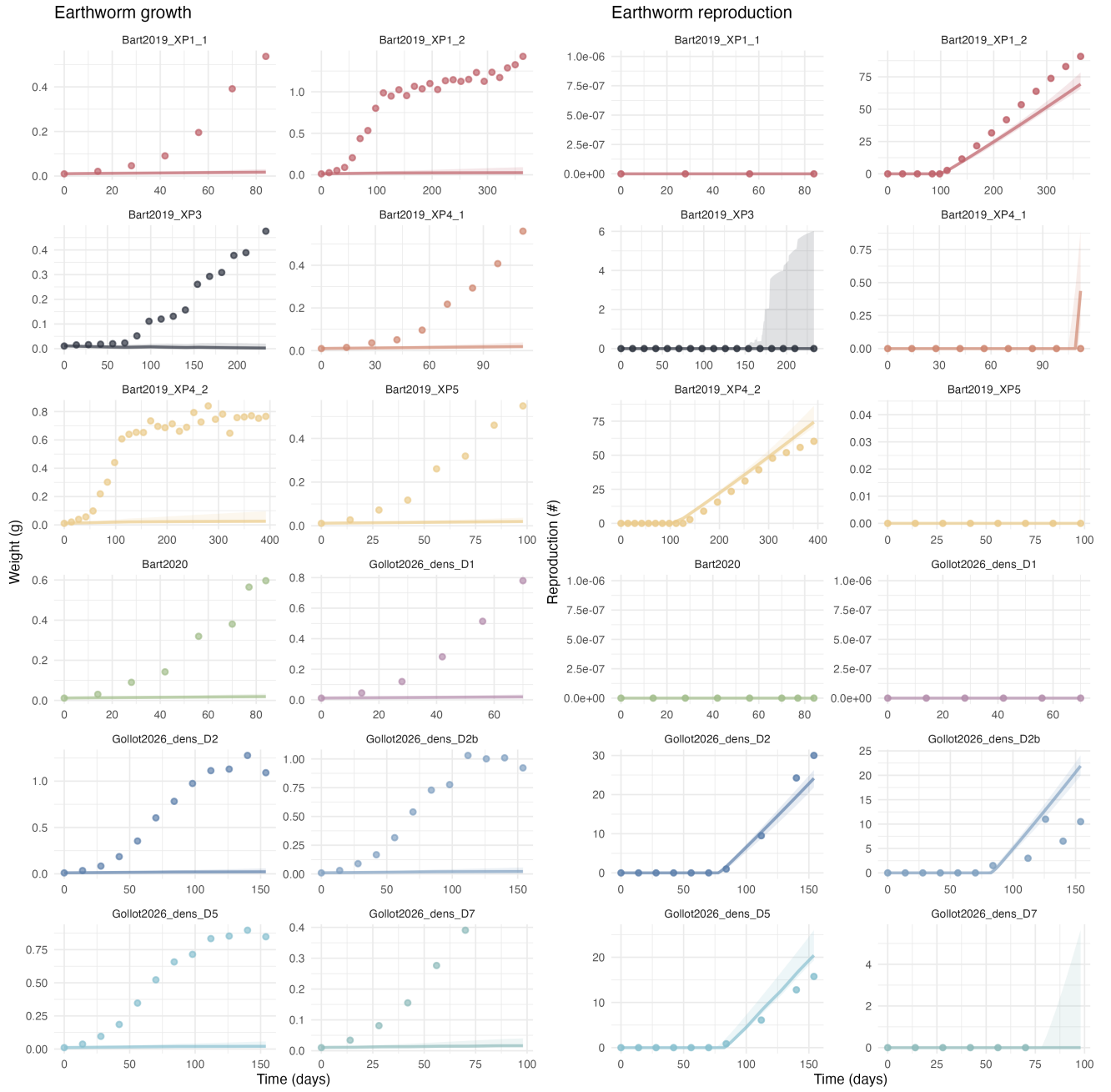


Figure 3.16: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

3.7 Model F_CLKe

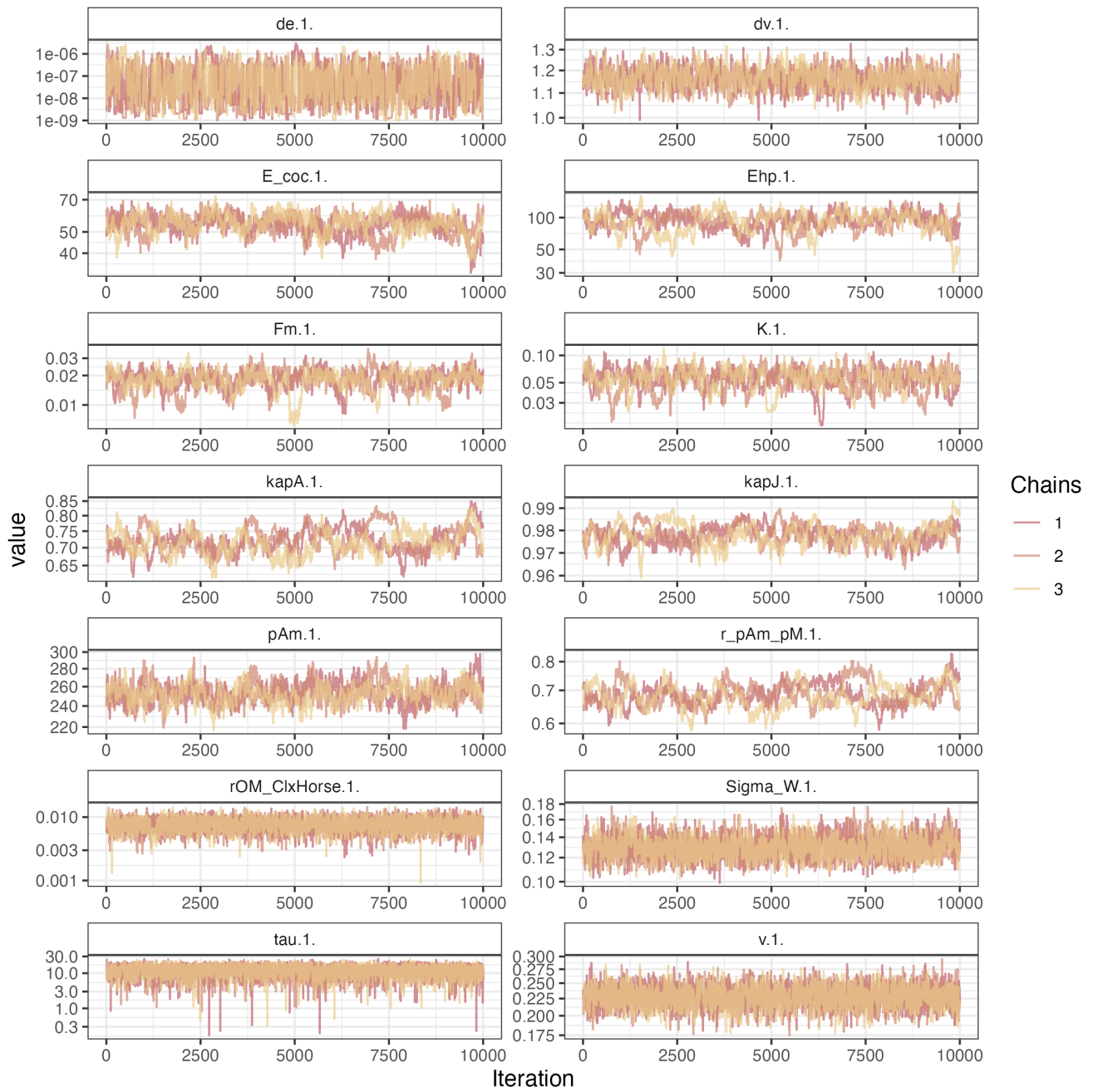


Figure 3.17: Chains.

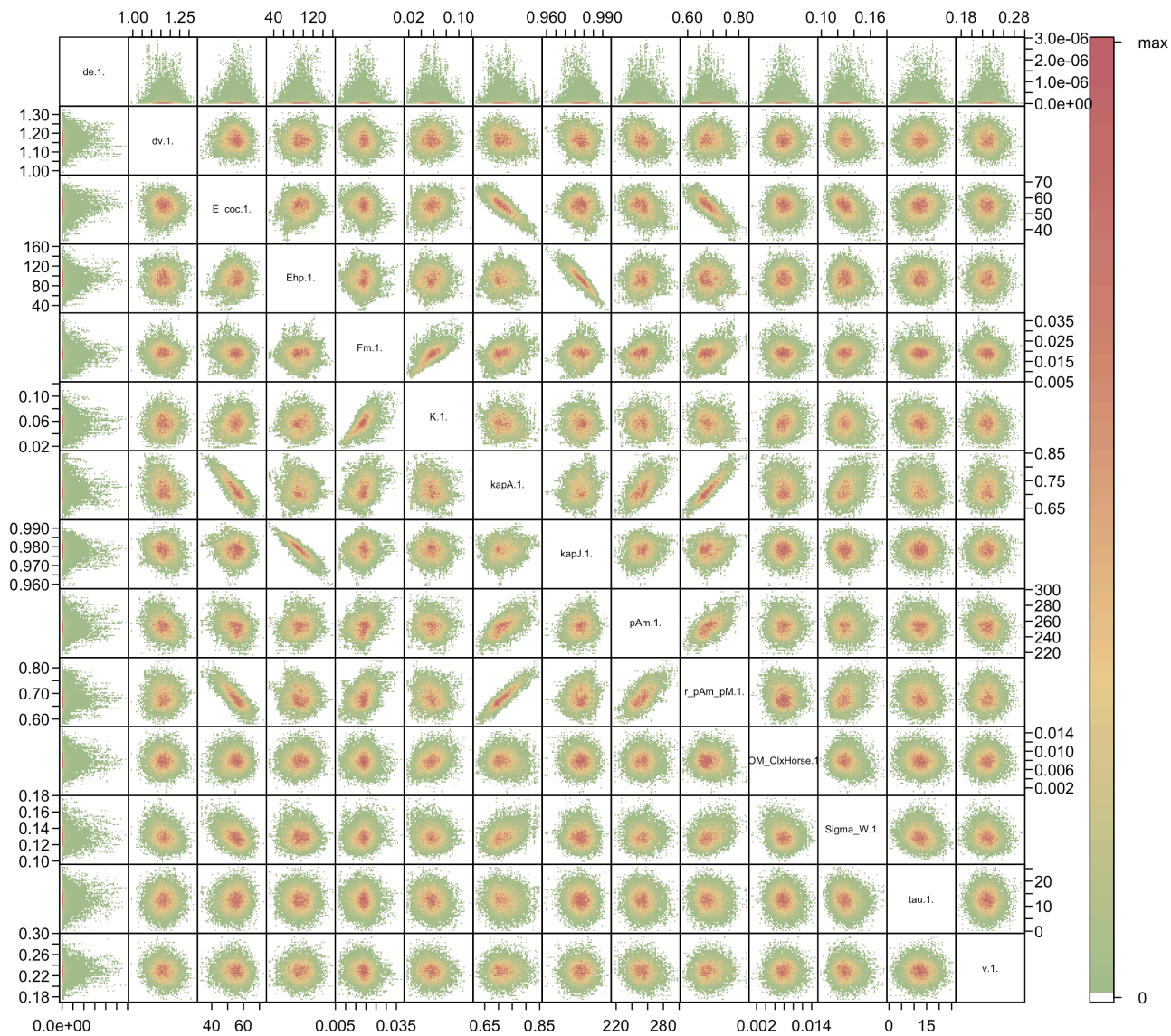


Figure 3.18: Chain correlations.

Posterior and prior distributions

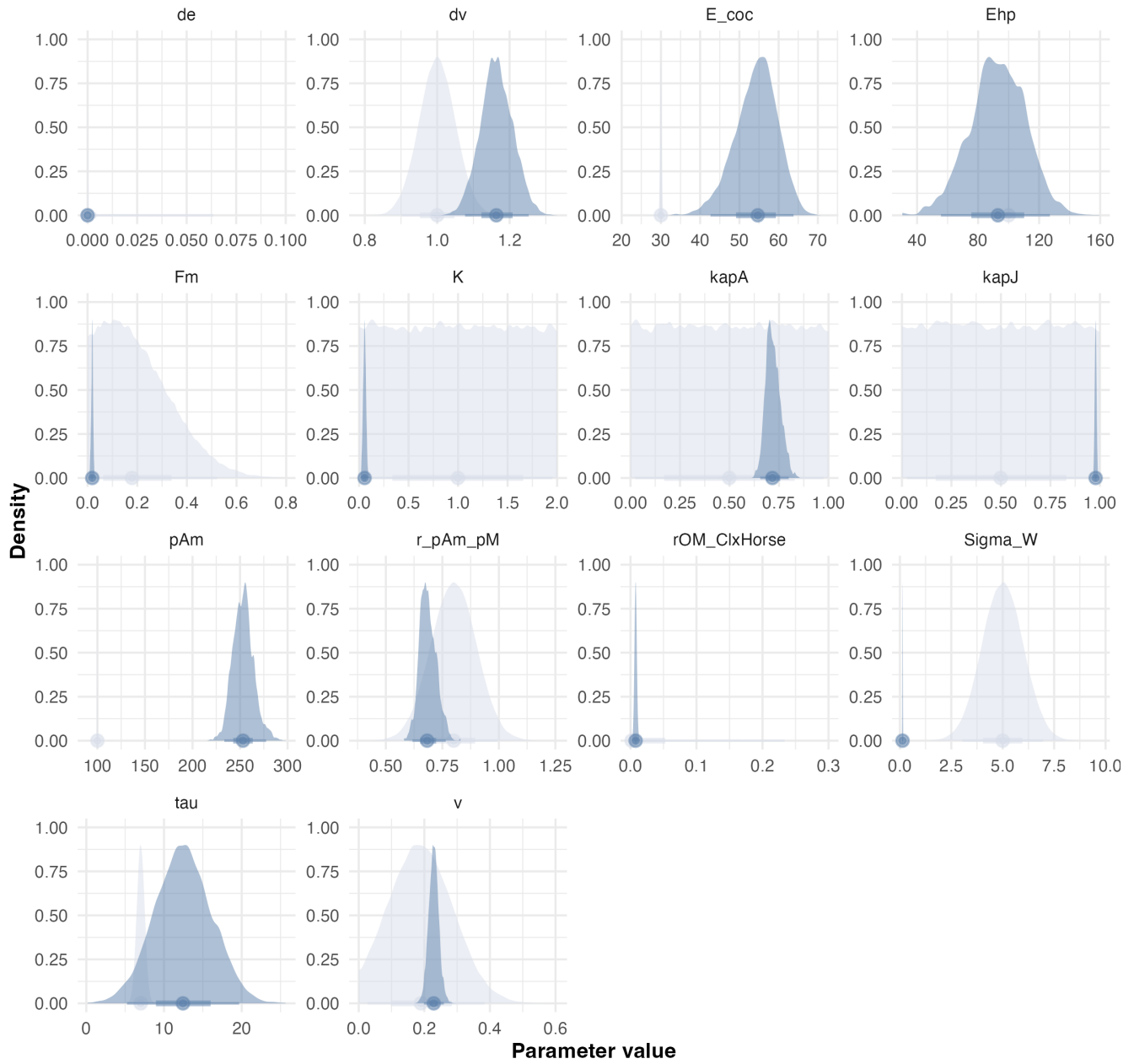


Figure 3.19: Posteriors.

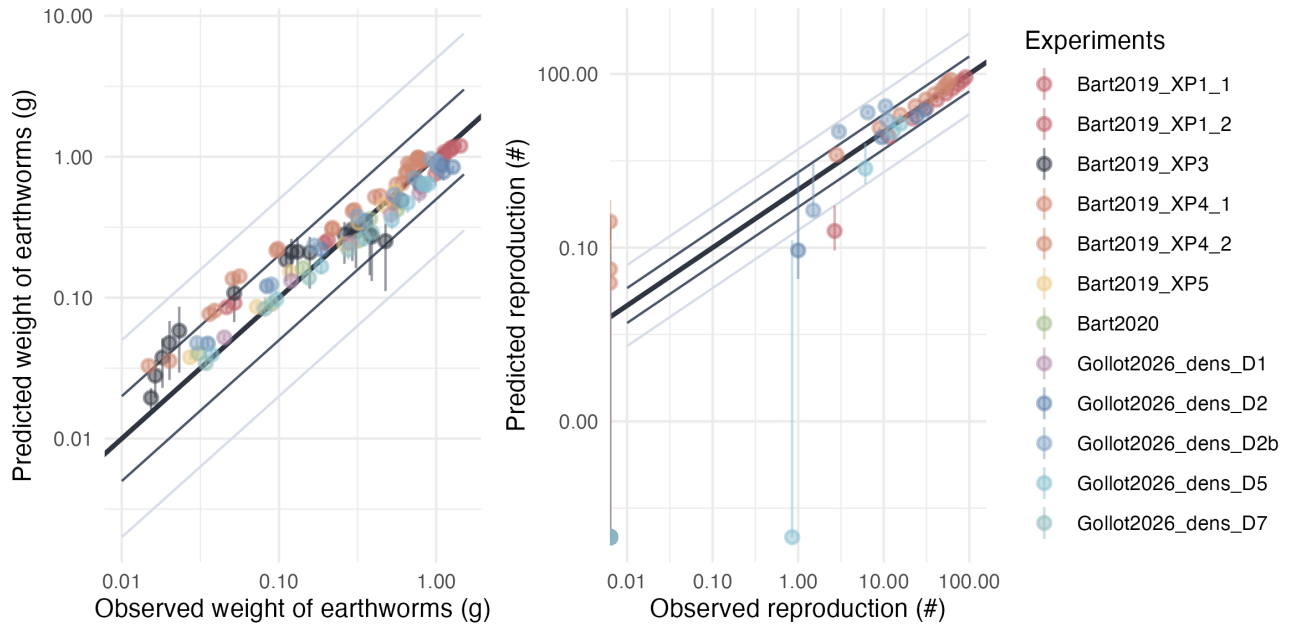


Figure 3.20: Predicted versus observed values of weight and reproduction. Points are colored by experimental conditions. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

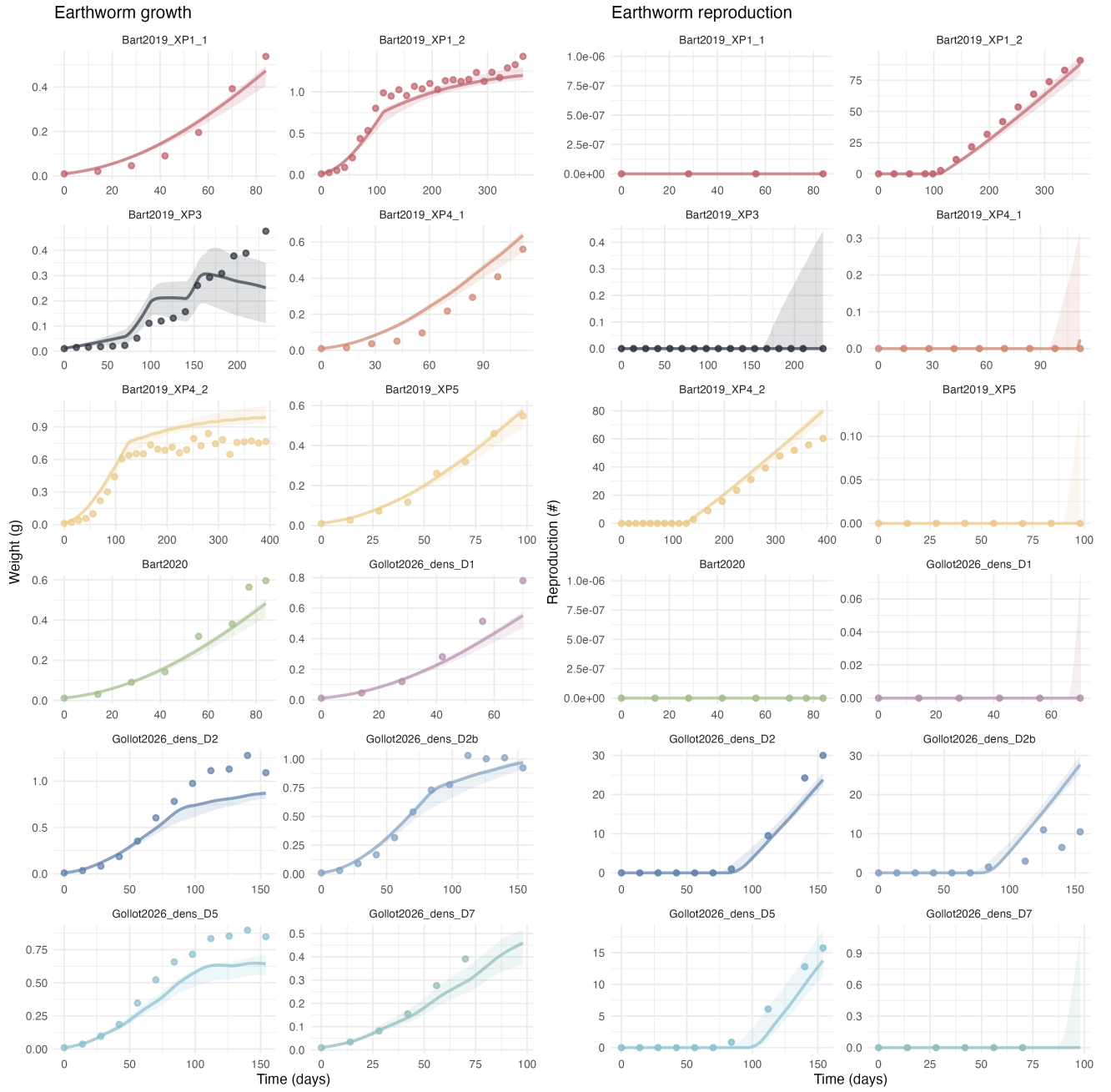


Figure 3.21: Observed and predicted growth and reproduction trajectories for each experiment . Points represent observed data, solid lines represent estimated model trajectories, and ribbons indicate the 95% credibility intervals.

3.8 Summary

Table 3.1: Estimate summary of each models and their respective log-likelihood and BIC.

Parameter	A_FLYw	B_FLYe	C_CWYe	D_CWKe	E_CLYe	F_CLKe
$\{\dot{p}_{Am}\}$	256.6 [240.6 ; 276.3]	281.6 [230.0 ; 316.8]	291.3 [258.0 ; 353.0]	254.5 [230.0 ; 276.6]	111.1 [104.7 ; 175.1]	254.7 [233.5 ; 277.4]
$[\dot{p}_M]/\{\dot{p}_{Am}\}$ ratio	1.09 [0.99 ; 1.13]	0.17 [0.12 ; 0.19]	0.76 [0.65 ; 0.79]	0.755 [0.617 ; 0.896]	0.67 [0.43 ; 0.74]	0.68 [0.62 ; 0.76]
$\{\dot{F}_m\}$ or $[\dot{F}_m]$	0.17 [0.12 ; 0.47]	0.13 [0.11 ; 0.28]	0.38 [0.30 ; 0.64]	0.019 [0.005 ; 0.033]	0.05 [0.05 ; 0.48]	0.019 [0.010 ; 0.026]
r_{OM}	0.011 [0.004 ; 0.015]	0.004 [0.004 ; 0.012]	0.01 [0.007 ; 0.019]	0.005 [0.002 ; 0.011]	0.0004 [0.0003 ; 0.012]	0.007 [0.004 ; 0.011]
K	NA	NA	NA	0.014 [0.003 ; 0.035]	NA	0.05 [0.03 ; 0.08]
v	0.20 [0.16 ; 0.23]	0.19 [0.16 ; 0.23]	0.21 [0.17 ; 0.24]	0.22 [0.20 ; 0.26]	0.18 [0.15 ; 0.22]	0.23 [0.20 ; 0.26]
κ	0.63 [0.52 ; 0.74]	0.92 [0.86 ; 0.94]	NA	NA	NA	NA
κ_J	NA	NA	0.94 [0.91 ; 0.98]	0.98 [0.97 ; 0.99]	0.74 [0.60 ; 0.957]	0.98 [0.97 ; 0.99]
κ_A	NA	NA	0.60 [0.55 ; 0.77]	0.77 [0.66 ; 0.81]	0.61 [0.49 ; 0.79]	0.71 [0.65 ; 0.80]
τ	NA	NA	7.1 [1.8 ; 15.3]	13.3 [6.4 ; 20.5]	6.8 [1.5 ; 15.0]	14.4 [5.3 ; 19.7]
E_h^p	24.5 [21.2 ; 29.0]	64.9 [41.5 ; 105.4]	52.1 [23.9 ; 87.7]	78.8 [51.2 ; 128.5]	36.5 [21.0 ; 73.8]	103.5 [55.6 ; 127.0]
E_{coc}	1.0 [1.0 ; 1.2]	4.7 [2.8 ; 7.6]	24.0 [12.5 ; 34.4]	46.9 [40.9 ; 63.7]	1.2 [1.0 ; 2.7]	56.2 [42.6 ; 63.8]

Parameter	A_FLYw	B_FLYe	C_CWYe	D_CWKe	E_CLYe	F_CLKe
d_V	1.0 [0.9 ; 1.1]	1.1 [0.94 ; 1.2]	1.1 [1.0 ; 1.2]	1.1 [1.1 ; 1.2]	1.0 [0.9 ; 1.1]	1.2 [1.1 ; 1.3]
d_E	NA	1.1e-9 [1.1e-9 ; 5.3e-7]	1.0e-9 [1.2e-9 ; 3.4e-7]	1.7e-9 [1.2e-9 ; 8.4e-6]	1.0e-9 [1.4e-9 ; 7.9e-6]	1.6e-9 [1.2e-9 ; 1.1e-6]
w_E	467.9 [249.3 ; 735.1]	NA	NA	NA	NA	NA
σ_W	0.16 [0.14 ; 0.19]	0.35 [0.31 ; 0.41]	0.32 [0.26 ; 0.37]	0.15 [0.12 ; 0.17]	0.64 [0.54 ; 0.69]	0.12 [0.11 ; 0.15]
Log-likelihood	-77.5	-199.1	-171.8	-55.8	-266.1	-38.0
BIC	211.2	454.5	411.1	184.7	599.7	149.2



Figure 3.22: RMSE for Weight and Reproduction per experiment for each tested model.

- Bart, S. (2019). Towards a better understanding of the life cycle of the earthworm *Aporrectodea caliginosa*: New data and energy-based modelling. *Pedobiologia*, 77, 150592. <https://doi.org/10.1016/j.pedobi.2019.150592>
- Bart, S., Laurent, C., Péry, A. R. R., Mougin, C., & Pelosi, C. (2017). Differences in sensitivity between earthworms and enchytraeids exposed to two commercial fungicides. *Ecotoxicology and Environmental Safety*, 140, 177–184. <https://doi.org/10.1016/j.ecoenv.2017.02.052>
- Bart, S., Pelosi, C., Nélieu, S., Lamy, I., & Péry, A. R. R. (2020). An energy-based model to analyze growth data of earthworms exposed to two fungicides. *Environmental Science and Pollution Research*, 27(1), 741–750. <https://doi.org/10.1007/s11356-019-06985-z>
- Kabus, J., Hartmann, V., Cocchiararo, B., Dombrowski, A., Enns, D., Karaouzas, I., Lipkowski, K., Pelikan, L., Shumka, S., Soose, L., Baker, N. J., & Jourdan, J. (2024). Cryptic species complex shows population-dependent, rather than lineage-dependent tolerance to a neonicotinoid. *Environmental Pollution*, 362, 124888. <https://doi.org/10.1016/j.envpol.2024.124888>